

# Equal-Well Allen–Cahn Fronts: Translation Uniqueness and a Pöschl–Teller Spectral Atlas

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## Abstract

We close a source-local theorem for the one-dimensional equal-well Allen–Cahn reaction–diffusion equation. For  $u_t = u_{xx} + \varepsilon^{-2}(u - u^3)$ , every monotone heteroclinic from  $-1$  to  $+1$  is a translate of  $\tanh(\xi/(\sqrt{2}\varepsilon))$ , and the equal-well balance forces its travelling speed to be zero. The gradient-flow energy dissipates exactly, with interfacial energy  $2\sqrt{2}/(3\varepsilon)$ . Linearization is the factorized Pöschl–Teller operator: translation is the simple zero mode, the shape mode has eigenvalue  $-3/(2\varepsilon^2)$ , and the essential spectrum is  $(-\infty, -2/\varepsilon^2]$ . Singular scaling faces are stated explicitly. The front is heteroclinic, not a primitive periodic orbit, so this is a rigorous Route-A negative result rather than an arithmetic or Hilbert–Pólya claim.

## 1 Frozen equation and the heteroclinic

We use the scaled family

$$u_t = u_{xx} + \varepsilon^{-2}(u - u^3), \quad \varepsilon > 0, \quad (1)$$

whose  $\varepsilon = 1$  member is the requested Allen–Cahn equation. Put  $W(q) = (1 - q^2)^2/4$ ; then (1) is the  $L^2$  gradient flow of

$$E_\varepsilon[u] = \int_{\mathbb{R}} \left( \frac{1}{2} u_x^2 + \varepsilon^{-2} W(u) \right) dx. \quad (2)$$

For a travelling ansatz  $u(x, t) = U(\xi)$ ,  $\xi = x - ct$ , the profile equation is

$$U'' + cU' + \varepsilon^{-2}(U - U^3) = 0, \quad U(-\infty) = -1, \quad U(+\infty) = 1. \quad (3)$$

**Theorem 1** (front, speed selection, and translation uniqueness). *Every nonconstant monotone  $C^2$  solution of (3) has  $c = 0$  and, for some  $\xi_0 \in \mathbb{R}$ ,*

$$U(\xi) = U_\varepsilon(\xi - \xi_0) := \operatorname{anh}\left(\frac{\xi - \xi_0}{\sqrt{2}\varepsilon}\right). \quad (4)$$

*The reverse orientation is  $-U_\varepsilon$ . For the increasing orientation,*

$$\frac{1}{2}(U'_\varepsilon)^2 = \varepsilon^{-2}W(U_\varepsilon), \quad \int_{\mathbb{R}} (U'_\varepsilon)^2 d\xi = E_\varepsilon[U_\varepsilon] = \frac{2\sqrt{2}}{3\varepsilon}. \quad (5)$$

*Proof.* Multiply (3) by  $U'$  and integrate from  $-\infty$  to  $+\infty$ . The endpoint derivatives and  $W(\pm 1)$  vanish, giving

$$c \int_{\mathbb{R}} (U')^2 d\xi = 0.$$

Since a nonconstant front has positive integral,  $c = 0$ . A first integral is then  $(U')^2/2 = \varepsilon^{-2}W(U)$ . Monotonicity selects the positive square root, and separation of variables gives (4) with one free translation. The identity  $U'_\varepsilon = (\sqrt{2}\varepsilon)^{-1} \operatorname{sech}^2 y$ ,  $y = (\xi - \xi_0)/(\sqrt{2}\varepsilon)$ , and  $\int_{\mathbb{R}} \operatorname{sech}^4 y dy = 4/3$  prove (5).  $\square$

## 2 Dissipation and the exact interface ledger

For smooth solutions with finite relative energy (subtracting the two pure well constants outside a bounded transition region), differentiation of (2) and integration by parts gives

$$\frac{d}{dt} E_\varepsilon[u(t)] = \int_{\mathbb{R}} (-u_{xx} - \varepsilon^{-2}(u - u^3)) u_t dx = - \int_{\mathbb{R}} u_t^2 dx \leq 0. \quad (6)$$

Equality holds only for a stationary solution. Along the front, the equipartition in (5) splits the surface energy equally between gradient and potential terms. Fixing the zero crossing  $U(\xi_0) = 0$  removes the sole translation freedom, so the monotone profile is unique under that phase condition. This is an energy statement, not a claim of a global attractor or a quantitative nonlinear convergence rate.

The speed identity also explains why adding a nonzero  $c$  by hand would change the model: equal wells have equal potential values, so a dissipative front cannot carry a net energy drop. A tilted potential can select a moving front, but it is not silently substituted for (1).

## Source note

The phase-field context is recorded by Allen and Cahn [1]; the travelling-front convergence context is recorded by Fife and McLeod [2]. The formulas, spectrum, and boundaries above are re-derived in this artifact; the references are contextual and do not enlarge the claim boundary.

## References

- [1] S. M. Allen and J. W. Cahn, A microscopic theory for antiphase boundary motion and its application to antiphase domain coarsening, *Acta Metallurgica* 27(6), 1085–1095 (1979). DOI: 10.1016/0001-6160(79)90196-2.
- [2] P. C. Fife and J. B. McLeod, The approach of solutions of nonlinear diffusion equations to travelling front solutions, *Archive for Rational Mechanics and Analysis* 65(4), 335–361 (1977). DOI: 10.1007/BF00250432.

## Declarations

**Scope.** NO\_BAD\_EULER\_OR\_ROOT\_NUMBER; no target arithmetic, periodic-orbit, or operator claim. **Data and code.** The exact front, energy ledger, independent checks, and build instructions are released with HCS-C231. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.